

Milky Way foreground-screen stratification in a SPARC/HECATE residual audit

Abstract

The Milky Way foreground is an observational layer, not merely a nuisance. Zone of Avoidance work with JWST/NIRCam has recently shown that faint background galaxies can be recovered through heavily contaminated Galactic regions, while also emphasizing that extinction, stellar crowding, and confusion noise shape extragalactic visibility. Motivated by this, we define a deliberately simple foreground-screen diagnostic for the public SPARC residual-disturbance audit: Galactic latitude computed from HECATE RA/DEC for exact-name SPARC overlaps. Using the provisional screen $|b| \leq 24^\circ$, the packet selects 18 systems. In the reviewed A/C subset, this screen is C-dominant: A=1 and C=6, with 11 ambiguous B systems. The A/C enrichment check is not statistically decisive ($p = 0.295343139746$), so the result is best read as an observer-screen audit and preregistration target, not a detection.

1. Motivation

Nilo-Castellon et al. identify 102 galaxies in a JWST/NIRCam field in the Zone of Avoidance and describe Galactic extinction, stellar crowding, and confusion noise as historical limits on background-galaxy detection. This motivates a conservative question for SPARC residual audits: do systems projected nearer to the Milky Way plane occupy a visibly different label/observability stratum?

The point of this note is not to explain galaxy dynamics. It is to preserve a small, reproducible foreground-screen packet that can be tested later with better extinction maps, star-count fields, and independent galaxy samples.

2. Question

The working question is deliberately modest:

When SPARC galaxies are stratified by a Milky Way foreground-screen proxy, does the low-latitude stratum show a distinctive label or residual pattern that

should be frozen as a future observer-screen test?

The note separates three claims that are often too easily mixed:

- a foreground-screen count claim;
- a residual-stress diagnostic;
- a physical interpretation.

Only the first two are tested here. The third is explicitly blocked.

3. Data And Construction

Inputs:

- HECATE v1.1 RA/DEC for exact-name SPARC overlaps.
- Residual-blind A/B/C labels from the public Paper 1 SPARC residual-disturbance packet.
- Public residual summary fields from the Paper 1 reproducibility package.

The script converts HECATE equatorial coordinates to Galactic longitude and latitude using the standard J2000 north Galactic pole constants. The primary screen is:

`low_latitude_screen = |b| <= 24 deg`

This threshold is not claimed as fundamental. It is a transparent reconstruction of the remembered 18-object screen and should be stress-tested in future work.

4. Foreground-Screen Diagnostic

Metric	Value	Interpretation
hecate_sparc_overlap_rows	120	exact-name SPARC/HECATE overlaps with RA/DEC
low_latitude_threshold_deg	24.0	preliminary foreground-screen proxy
low_latitude_total	18	b <= 24 deg
low_latitude_A	1	reviewed calm/regular class count
low_latitude_B	11	ambiguous class count
low_latitude_C	6	reviewed disturbed class count
high_latitude_A	15	b > 24 deg
high_latitude_B	57	b > 24 deg
high_latitude_C	30	b > 24 deg
ac_one_sided_c_enrichment_p	0.295343139746	diagnostic only; not a detection

Threshold Scan

AbsGalacticLatitudeThresholdDeg	N_total	N_AC	N_A	N_B	N_C
5	0	0	0	0	0
7.5	0	0	0	0	0

10	1	0	0	1	0
12	2	0	0	2	0
15	8	2	0	6	2
20	13	5	0	8	5
22	15	6	1	9	5
23	17	6	1	11	5
24	18	7	1	11	6
24.5	18	7	1	11	6
25	19	8	1	11	7
26	20	9	1	11	8
27	21	10	2	11	8
28	21	10	2	11	8
29	23	12	2	11	10
30	25	13	3	12	10

5. The 18 Low-Latitude Systems

GalaxyName	Class	AbsGalacticLatitudeDeg	GalacticLongitudeDeg	HecateDistanceMpc	RmsLogTPG	MeanLogResidualTPG	OuterMeanLogResidualTPG
UGC03205	B	8.206377	342.903037	54.37	0.11600949975788537	0.0873983245850471	0.055565320187085834
NGC6946	B	11.672188	60.144887	5.516	0.0896359090075854	-0.09256925858478726	-0.0831693606116379
UGC11557	B	12.814719	60.883601	23.66	0.7431151375673428	-0.6040787203825251	-0.35763864335701157
UGC00731	C	13.149452	29.723502	13.03	0.339703437495753	-0.0782044521386274	0.21001308628006135
NGC6674	B	13.919223	101.271460	54.11	0.1261360715254176	0.10826931615214701	0.10598763897950496
UGC02885	B	14.050142	356.283096	71.12	0.06662315534970306	0.04685592336981314	0.0463394047425374
UGC02916	C	14.194665	19.057928	66.11	0.13928911042196776	-0.06831440677751897	-0.07903890648018923
ESO563-G021	B	14.397417	270.996728	79.44	0.29404072405040915	0.07931689655380693	0.1691008384002877
NGC0891	C	17.415249	15.479606	9.111	0.1259442118714204	-0.09347531570220223	-0.05008617779939154
NGC1003	B	17.545110	11.859346	10.76	0.12113411141685551	0.024974111717001073	0.14932873924545975
NGC2915	C	18.357173	223.897427	4.03	0.44092390599176534	0.27527781688707265	0.3774115856001438
UGC12632	C	19.311054	49.092790	9.204	0.1052568716207706	-0.07054854024451106	0.0485761511506591
UGC02259	B	19.796208	8.718768	9.954	0.26943439551068704	0.24286674478824713	0.15148615161698695
NGC7331	A	20.724275	62.141728	14.4	0.09804264249274548	-0.08633593090910359	-0.03659801898184237
NGC6789	B	21.517771	60.889695	3.267	0.3895854018308549	0.5020305791744587	0.35690932806891795
NGC0801	B	22.456867	17.644924	55.53	0.1408128804442707	-0.08465940603010062	-0.17407443093346248
UGC11455	B	22.844922	52.115383	85.75	0.14196092541106067	-0.06123895742954915	-0.028053098839149467
NGC3109	C	23.070239	253.762265	1.291	0.17587234171290073	0.1284165505065754	0.24323203399945445

6. Label-Stratum Interpretation

The result is directionally interesting because the low-latitude screen is

C-heavy among systems that already have reviewed A/C labels. However, the sample

is small and B-dominated. A one-sided A/C hypergeometric enrichment diagnostic

returns $p = 0.295343139746$, which is not a discovery-level result.

7. Residual-Signal Stress Test

The natural follow-up question is whether the 18-object screen is also a good place to look for a residual projection signal. We therefore join the foreground screen to the public Paper 1 residual summary and test two distinct possibilities:

1. A residual-scatter excess.
2. A signed residual offset.

The current packet does **not** support a simple RMS-excess claim. The median ``rms_log_tpg`` in the low-latitude screen is slightly lower than the high-latitude comparison set. The more interesting candidate is instead a signed offset: ``mean_log_residual_tpg`` and ``outer_mean_log_residual_tpg`` shift positive in the low-latitude screen. This is exactly the sort of weak observer-screen candidate that should be preregistered before any stronger interpretation.

Metric	Value	Interpretation
residual_summary_available	true	public Paper 1 residual summary joined to foreground screen
residual_join_rows	120	HECATE/SPARC rows with residual summary
residual_low_screen_rows	18	$b \leq 24$ deg
residual_high_screen_rows	102	$b > 24$ deg
rms_log_tpg_low_median	0.140050995433	low-latitude foreground screen
rms_log_tpg_high_median	0.146757312311	higher-latitude comparison set
rms_log_tpg_low_minus_high_median	-0.00670631687742	diagnostic difference; not a detection
weighted_rms_log_tpg_low_median	0.116555387387	low-latitude foreground screen
weighted_rms_log_tpg_high_median	0.137194953464	higher-latitude comparison set
weighted_rms_log_tpg_low_minus_high_median	-0.0206395660763	diagnostic difference; not a detection
mean_log_residual_tpg_low_median	0.0477160372602	low-latitude foreground screen
mean_log_residual_tpg_high_median	-0.039649763687	higher-latitude comparison set
mean_log_residual_tpg_low_minus_high_median	0.0873658009472	diagnostic difference; not a detection
outer_mean_log_residual_tpg_low_median	0.00914315295169	low-latitude foreground screen
outer_mean_log_residual_tpg_high_median	-0.0241318063717	higher-latitude comparison set
outer_mean_log_residual_tpg_low_minus_high_median	0.0332749593234	diagnostic difference; not a detection
mean_err_vobs_kms_low_median	5.77818965517	low-latitude foreground screen
mean_err_vobs_kms_high_median	5.1711037234	higher-latitude comparison set
mean_err_vobs_kms_low_minus_high_median	0.607085931768	diagnostic difference; not a detection
max_radius_kpc_low_median	25.32	low-latitude foreground screen
max_radius_kpc_high_median	12.935	higher-latitude comparison set
max_radius_kpc_low_minus_high_median	12.385	diagnostic difference; not a detection
n_points_low_median	22	low-latitude foreground screen
n_points_high_median	13	higher-latitude comparison set
n_points_low_minus_high_median	9	diagnostic difference; not a detection

interpretation	candidate_signed_projection_offset_not_rms_excess_detection	RMS
median is not elevated in the 18-object screen; signed residual medians shift positive		

8. Matched-Control Preview

As a first guardrail, each low-latitude galaxy is greedily matched to a unique higher-latitude control by log HECATE distance, log radial extent, and log point count. This is only a preview; it is not a covariance-aware likelihood or an extinction-aware analysis.

Metric	Value	Interpretation
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matched_pair_count	18	greedy unique high-latitude controls matched on log distance, max radius, and point count
matched_rms_log_tpg_median_low_minus_high	-0.00641394691747	paired diagnostic difference; not a detection
matched_rms_log_tpg_positive_pairs	9/18	sign count across matched pairs
matched_weighted_rms_log_tpg_median_low_minus_high	0.0158783211739	paired diagnostic difference; not a detection
matched_weighted_rms_log_tpg_positive_pairs	10/18	sign count across matched pairs
matched_mean_log_residual_tpg_median_low_minus_high	0.0100950616694	paired diagnostic difference; not a detection
matched_mean_log_residual_tpg_positive_pairs	9/18	sign count across matched pairs
matched_outer_mean_log_residual_tpg_median_low_minus_high	0.0105481845933	paired diagnostic difference; not a detection
matched_outer_mean_log_residual_tpg_positive_pairs	9/18	sign count across matched pairs

9. What The Result Does Not Say

The current packet does not show that the Milky Way foreground creates galaxy disturbance. It also does not show that a new dynamical law is required. The positive signed residual shift may come from ordinary observational effects: foreground extinction, stellar crowding, different target selection, distance and radius imbalance, inclination systematics, or residual calibration choices.

The result is useful because it defines a clean future test. If a foreground observer-screen effect is real, it should survive after replacing ``|b|`` with physical foreground maps such as ``E(B-V)``, ``A_V``, source density, and confusion metrics. If it disappears under those controls, the present 18-object window was only a selection artifact.

The correct reading is therefore:

The Milky Way foreground screen is a real observability stratum. In this SPARC/HECATE packet, a reconstructed 18-object low-`|b|` screen is C-dominant among

reviewed A/C labels, but the result remains a preregistered audit target rather than a physical inference.

The residual follow-up does not show a robust RMS-excess detection. It does show a candidate signed residual-offset direction that is worth freezing as the next observer-screen test.

10. Preregistered Next Tests

1. Replace the latitude-only proxy with $E(B-V)$, A_V , star-count density, and local confusion metrics.
2. Freeze the screen thresholds before residual inspection.
3. Re-run the audit on independent low-latitude and high-latitude galaxy samples.
4. Separate foreground observability effects from intrinsic disturbance evidence.
5. Report all A/B/C transitions under threshold scans rather than selecting only the most favorable boundary.
6. Test signed residual offset before RMS excess, because the present packet disfavors the simplest RMS-excess version of the hypothesis.

11. Claim Boundary

This note does not claim new dynamics, a physical detection, or proof of any private parent theory. It is a public observer-screen method note.

References

- J. L. Nilo-Castellon et al., *Faint galaxies in the Zone of Avoidance revealed by JWST/NIRCam*, arXiv:2510.12488, A&A; 704, A209 (2025).
- SPARC residual-disturbance Paper 1 public reproducibility package.
- HECATE v1.1 public catalog.